

# Thevenin's Theorem Norton's theorem.

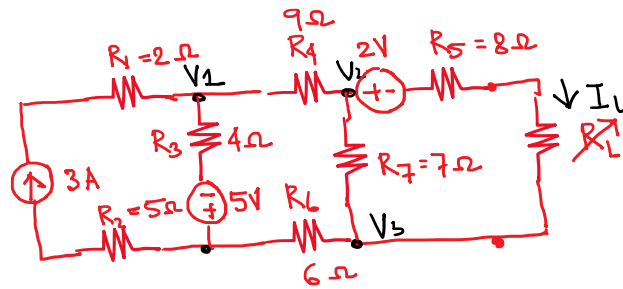
$R_L$ : load Resistance

$R_L = 10\Omega, 15\Omega, 20\Omega$

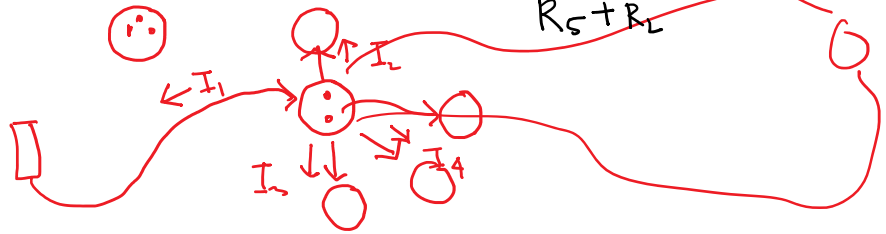
$I_L = ?$

$R_L = \text{phone, TV, Laptop, tab.}$

$I_c = I_c$

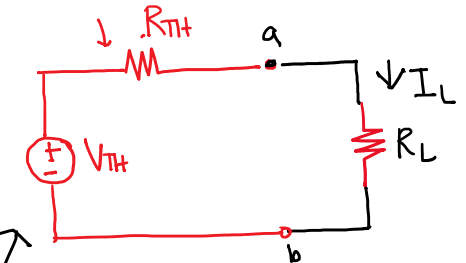


$$I_L = \frac{V_2 - 2 - V_3}{R_5 + R_L}$$



Thevenin's Voltage:  $V_{TH}$ .  $R_{TH}$ : Thevenin's Resistance

Part A



$R_L = 10\Omega, 20\Omega, 15$

$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

Part A

Part B

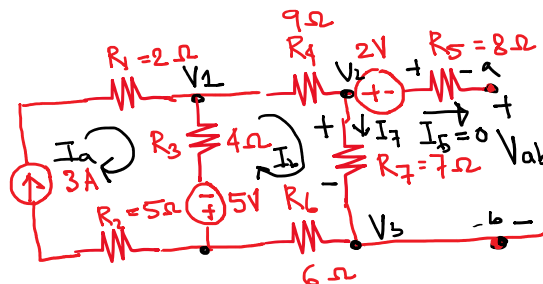
Disconnect Part-B from Part A : Part-B Removed

bet<sup>n</sup> Node a & b it's open

□ determine  $V_{ab} = V_{oc}$

$V_{oc}$ : open ckt voltage bet<sup>n</sup> a & b

$$V_{TH} = V_{oc} = V_{ab}$$



$$I_a = 3A$$

$$-5 - 4(I_b - I_a) - 9I_b - 7I_b - 6I_b = 0$$

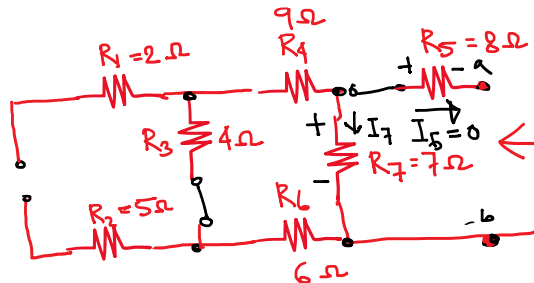
$$-4I_b + 12 + 22I_b = 5 \Rightarrow 26I_b = 7$$

$$+V_{R7} - 2 - V_{R5} - V_{ab} = 0$$

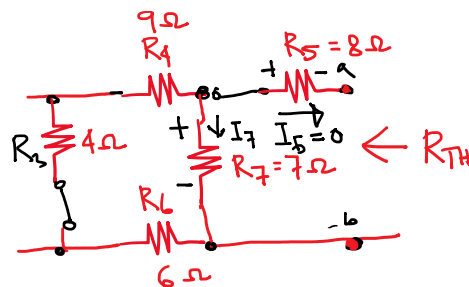
$$+7 \times I_b - 2 - 5I_5 - V_{ab} = 0 \quad I_7 = I_b = \frac{7}{21} = 0.269 \text{ A}$$

$$7 \times 0.269 - 2 - V_{ab} = 0 \quad V_{ab} = -0.117 \text{ V}$$

$$V_{TH} = V_{OC} = V_{ab} = -0.117 \text{ V}$$



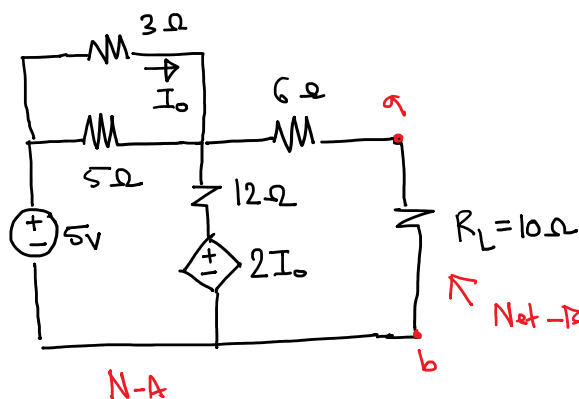
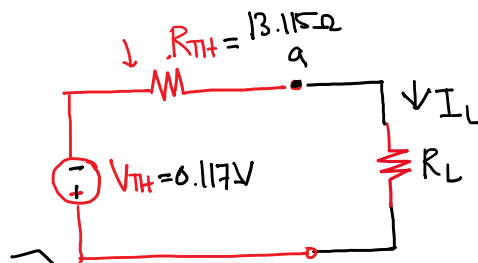
Ind. Current source is open,  
Ind. Voltage source is short



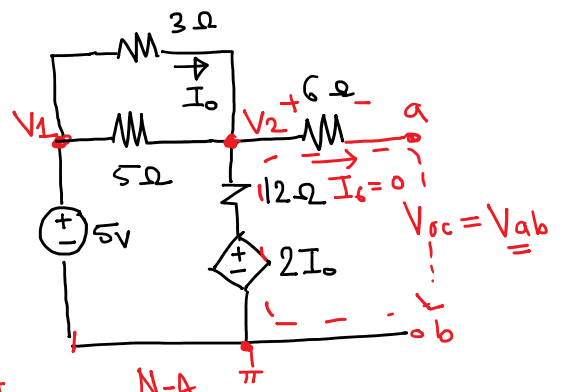
$$R_{TH} = [(R_4 + R_3 + R_6) \parallel R_7] + R_5$$

$$= (19 \Omega \parallel 7 \Omega) + 8 \Omega$$

$$= 5.115 + 8 = 13.115 \Omega$$



Disconnected Net-B



$$V_1 = 5 \text{ V}$$

$$\frac{V_2 - V_1}{3} + \frac{V_2 - V_1}{5} + \frac{V_2 - 2I_o}{12} = 0$$

$$\frac{V_1 - V_2}{3} = I_o$$

$$\frac{V_2 - 5}{3} + \frac{V_2 - 5}{5} + \frac{V_2 - 2(5 - V_2)}{12} = 0$$

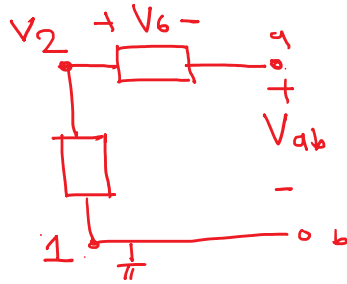
$$\dots \rightarrow V_1 =$$

$$V_2 = x_1$$

$$V_2 - V_1 - V_{ab} = 0$$

$$\frac{1}{3} = I_0 \quad |$$

$$\frac{1}{12} = I_0$$



$$V_2 = X_1$$

$$V_2 - V_6 - V_{ab} = 0$$

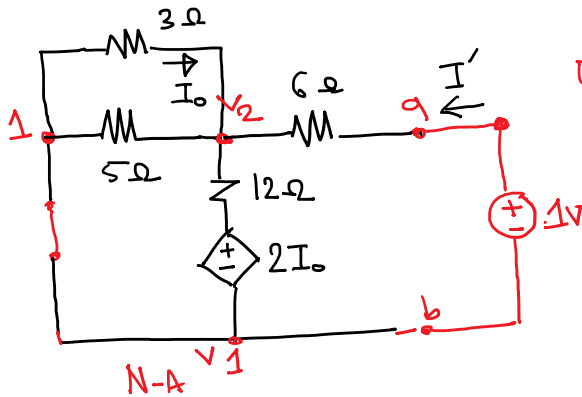
$$\Rightarrow X_1 - 6I_0 - V_{ab} = 0$$

$$\Rightarrow V_{ab} = X_1 = V_{oc} = V_{TH}$$

$$(V_2 - 0) - V_6 - V_{ab} = 0 \Rightarrow V_2 - V_6 - V_{ab} = 0$$

$$\Rightarrow X_1 - 6I_0 - V_{ab} = 0$$

$$V_{TH} = V_{ab} = X_1$$



- Remove Independent sources?
- dependent sources? don't Remove

$$R_{TH} = \frac{1V}{I'}$$

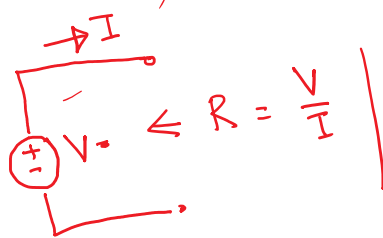
$$\frac{V_2 - 0}{5} + \frac{V_2 - 0}{3} + \frac{V_2 - 2I_0}{12} + \frac{V_2 - 1}{6} = 0$$

$$I_0 = \frac{V_1 - V_2}{3} = \frac{0 - V_2}{3} = -\frac{V_2}{3}$$

$$\frac{V_2}{5} + \frac{V_2}{3} + \frac{V_2 - 2(-\frac{V_2}{3})}{12} + \frac{V_2 - 1}{6} = 0$$

$$I' = \frac{1 - V_2}{6} = \frac{1 - X_2}{6}$$

$$V_2 = X_2$$



$$R_{TH} = \frac{1V}{I'}$$

$$R_{TH} = \frac{V_{ab}}{I'} = \frac{V_{ab}}{1A}$$

